

Skills Summary

Estimating Differences and Checking Answers

Use this reference while completing your worksheet or when studying.

Skill 1 — Estimating to the nearest ten

Rule: Round *both* numbers to the nearest ten, then subtract the rounded numbers.

Look at the ones digit: below five rounds down, five or more rounds up.

Never round only one of the two — an estimate built from one rounded number and one exact one is neither quick nor honest.

Example: Estimate $83 - 46$ to the nearest ten.

Step 1. I want a quick answer I can carry in my head, and tens are the easiest numbers to subtract, so I move both numbers to the nearest ten before doing anything else.

Step 2. The first number has three ones, so it rounds down to eighty. The second has six ones, so it rounds up to fifty. Eighty take away fifty is the estimate.

30

Try it: Estimate $71 - 34$ to the nearest ten.

check: 40

Skill 2 — Estimating to the nearest hundred

Rule: Same method, one place further left: round both numbers to the nearest hundred, then subtract.

Look at the *tens* digit this time: below five rounds down, five or more rounds up.

Hundreds give a rougher estimate than tens, but the subtraction becomes one you can do in your head.

Example: Estimate $573 - 241$ to the nearest hundred.

Step 1. These numbers are in the hundreds, so rounding to tens would still leave me subtracting awkward numbers; rounding to hundreds gives a subtraction I can say out loud.

Step 2. The first number has seven tens, so it rounds up to six hundred. The second has four tens, so it rounds down to two hundred. Six hundred take away two hundred is the estimate.

400

Try it: Estimate $418 - 262$ to the nearest hundred.

check: 100

Skill 3 — Checking an answer with an estimate

Rule: Estimate first, subtract exactly second, then compare the two.

Rounding can move an answer by a few tens. It cannot move it by a whole hundred — so a gap that big means the exact working has a mistake in it, not that the estimate was poor.

Example: Estimate $645 - 328$ to the nearest hundred, then subtract exactly and compare.

Step 1. The point of the estimate is to have something to test the exact answer against, so I work it out first and keep it beside me while I subtract.

Step 2. Rounding gives six hundred take away three hundred. Exactly: regroup for the ones, leaving six hundreds, three tens and fifteen ones; fifteen take eight leaves seven, three take two leaves one, six take three leaves three. The two answers land within a few tens of each other, so nothing was dropped.

estimate **300** exact **317**

Try it: Estimate $532 - 219$ to the nearest hundred, then subtract exactly.

check: estimate 300, exact 318

Skill 4 — Spotting an answer the estimate rules out

Rule: If an exact answer sits more than about a hundred away from a nearest-hundred estimate, do not accept it — go back and look for a lost regrouping.

Being close to the estimate does not prove an answer right, but being far from it does prove one wrong.

Example: Someone worked out $700 - 264$ and wrote 564. Is that reasonable?

Step 1. Rather than redo the whole subtraction straight away, I estimate first: if the written answer is nowhere near the estimate, I already know it is wrong and where to look.

Step 2. Rounding gives seven hundred take away three hundred, so about four hundred should be left. The written answer sits more than a hundred above that, which rounding cannot explain. Subtracting properly — six hundreds, nine tens, ten ones — gives the true difference.

estimate **400** correct **436**

Try it: Someone worked out $805 - 397$ and wrote 508. Estimate to the nearest hundred, then find the correct difference.

check: estimate 400, correct 408